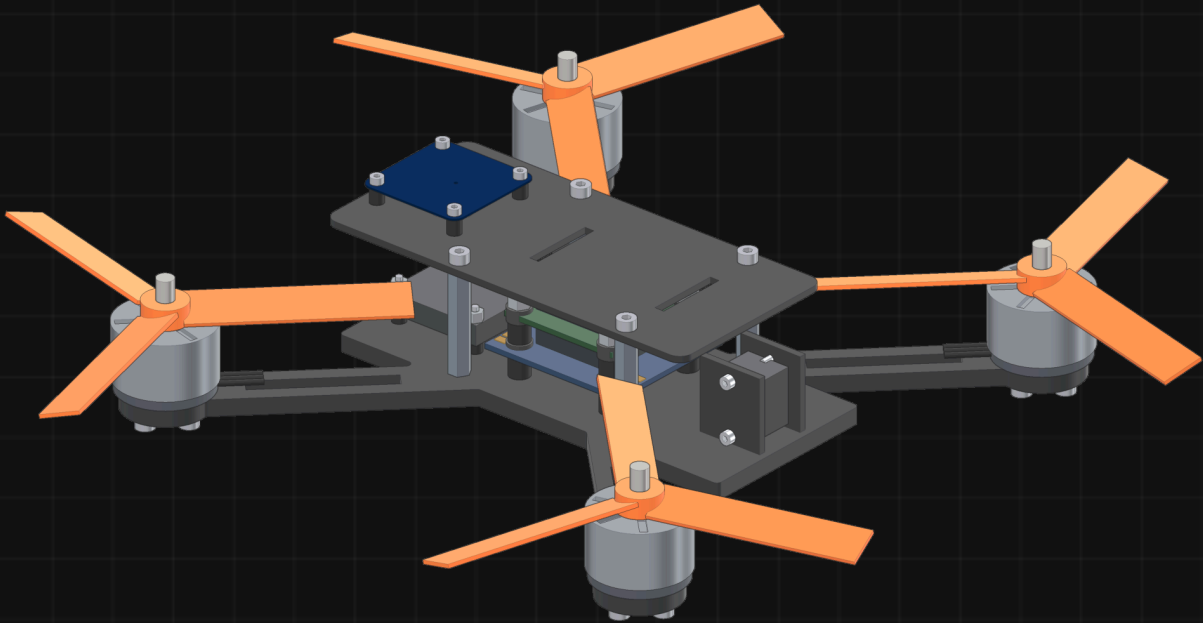


5-INCH 6S QUADCOPTER

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.



Assembled from UHD mates (16 placed bodies, 72 fasteners); strapped battery not shown.

STATIC THRUST 6.3 kg^D 4 motors, full throttle, maker's test prop	BATTERY 22.2 V 6S LiPo	CAPACITY 1100 mAh XT60
PEAK CURRENT 171.9 A 4 × motor 60 s peak	ALL-UP WEIGHT 584 g^A systemMass; cables and capacitor not included	VIDEO LATENCY 20 ms DJI O4, Goggles 3

■ ABOUT THIS DOCUMENT

Every value is queried from the UHD model at build time and re-checked by `uhd-tech-docs-verify`. Markers show where a value is not a cited source value.

- ^A Assumption: the model marks this value as an assumption (assumption trait)
 - ^D Derived: computed from model values; formula in the provenance appendix
 - ^S Stand-in: stand-in data derived from the model, not measured
 - Not in model: the model does not state this value (data gap)
- Unmarked values are cited from the part's sources (`sources.json`) or are design values of the model.

01 OVERVIEW

QUADCOPTER-SIN · V0.1.0

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.

ROLE	MODULE	QTY
Frame	5-inch quadcopter frame (custom)	1
Top plate	5-inch frame top plate (custom)	1
FC + ESC stack	DolphinRC F405 V3 + AM32 60A stack	1
Arms	Arm (motor + propeller)	4
6S battery	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	1
ELRS receiver	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	1
DJI O4 Air Unit	DJI O4 Air Unit	1
GNSS + compass	Matek M9N-5883 GNSS & Compass	1
Battery strap	HGLRC Kevlar battery strap 20x250mm	1
Bulk capacitor 470 µF	Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	1

CONTENTS

01	Overview	02
02	Specifications	03
03	Mechanical	04
04	Interfaces and pinout	07
05	Electrical characteristics	08
06	Performance	09
07	Design checks	11
08	Bill of materials	12
09	Evidence and assumptions	14
10	Value provenance	16

SYSTEM ARCHITECTURE

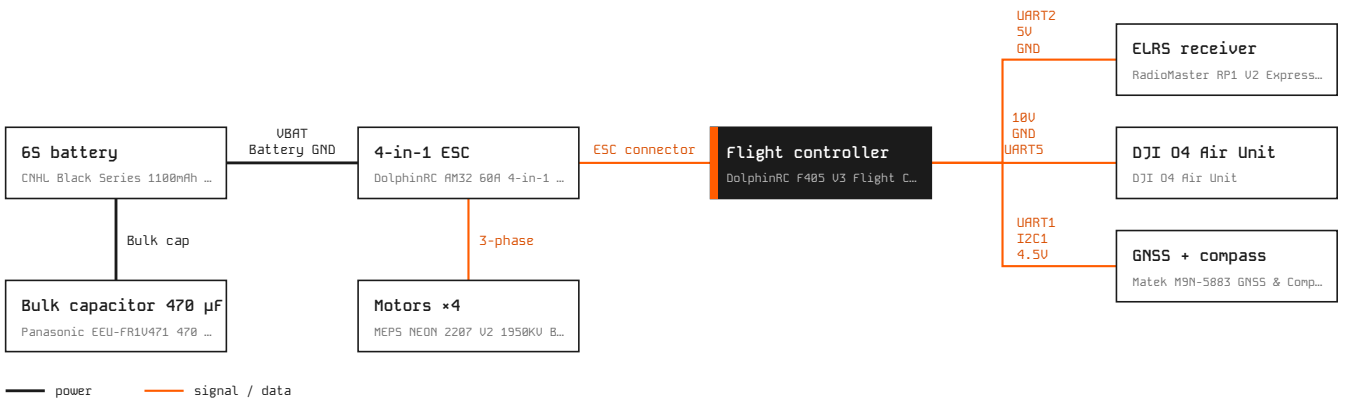


Figure 1 Power and signal links as stored in the model (validated with validateLinks); labels are the hub-side interfaces.

02 SPECIFICATIONS

PARAMETER	VALUE
AIRFRAME	
Layout	true-X, 5-inch
Wheelbase	225 mm
Bottom plate	5 mm
Standoff height	30 mm
Frame mass carbon, from CAD	91.0 g ^A
Stack pattern square, M3	30.5 mm
Motor pattern bolt circle, M3	16 mm
PROPULSION	
Motor	MEPS NEON 2207 V2 1950KV Brushless Motor
KV	1950 rpm/V
Stator / poles	12N14P
Peak current (60 s)	42.97 A
Max power	1036 W
Motor mass	36 g
Propeller	HQProp Ethix S5 5x4x3 Propeller
Prop mass	3.7 g
ESC per channel continuous	60 A
ESC burst	65 A
ESC firmware	AM32 (32-bit); firmware target AT32DEVF421
FLIGHT CONTROL	
Flight controller	DolphinRC F405 V3 Flight Controller
MCU	STM32F405RGT6, Cortex-M4 @ 168MHz
IMU	ICM-42688-P
Firmware target	Betaflight MATEKF405TE
UARTs	6
Receiver	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver
RC link	ExpressLRS
CRSF baud rate	420 000 Bd

PARAMETER	VALUE
VIDEO	
Air unit	DJI O4 Air Unit
Image sensor	1/2-inch CMOS
Field of view	117.6°
Max bitrate	100 Mbit/s
Supply range	3.7–13.2 V
Operating temperature	-10–40 °C
POWER	
Battery	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)
Nominal voltage	22.2 V
Capacity	1100 mAh
Energy	24.42 Wh ^D
Continuous current	110 A
Burst current	220 A
Pack mass	211 g
All-up weight excl. cables, cap	584.0 g ^A
ESC input	6–30 V
BEC 5 V rating	3 A
BEC 10 V rating	2.5 A
GPS rail	4.5–5.0 V
NAVIGATION	
GNSS module	Matek M9N-5883 GNSS & Compass
Receiver	u-blox NEO-M9N (concurrent reception of GPS, Galileo, GLONASS, BeiDou)
Compass address	0x2C
Supply current	50 mA
GPS rail fed from assumed	bec_5v ^A

^A Assumption: the model marks this value as an assumption (assumption trait)

^D Derived: computed from model values; formula in the provenance appendix

^S Stand-in: stand-in data derived from the model, not measured - Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

03 MECHANICAL

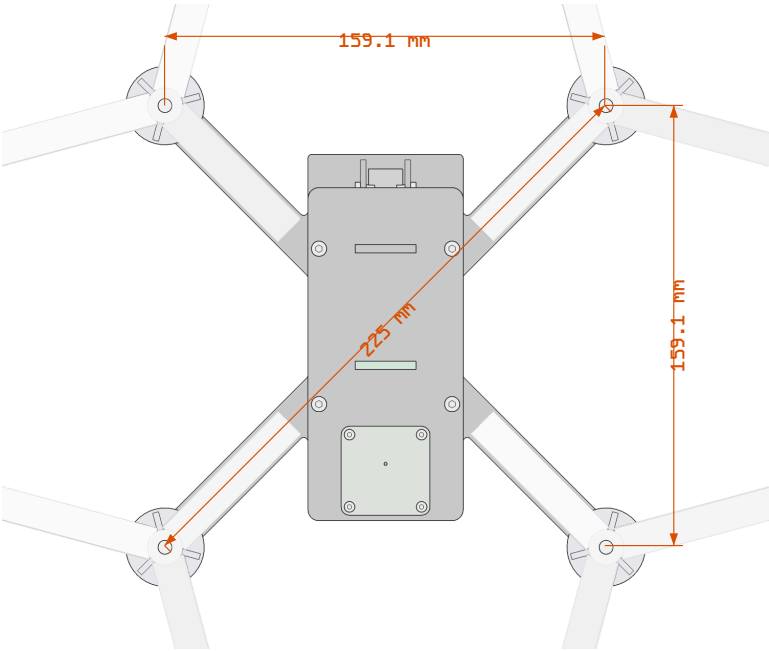


Figure 2 Top view, front up. Wheelbase is the frame's stated value; arm spacings are measured between the placed motor-mount frames. (unit: mm)

ENVELOPE

PARAMETER	VALUE
Wheelbase	225 mm
Arm spacing (x)	159.1 mm ^D
Arm spacing (y)	159.1 mm ^D
Plate thickness	5 mm
Standoff height	30 mm
Propeller diameter	5 in
Motor diameter	28.3 mm
Plate material	3K woven carbon fibre / epoxy laminate ^A
Material density	1.52 g/cm ³ ^A
Frame mass	91.0 g ^A
Top plate mass	19.8 g ^A

Plate masses are the generated CAD volume × the declared material density (systemMass), so they follow the design.

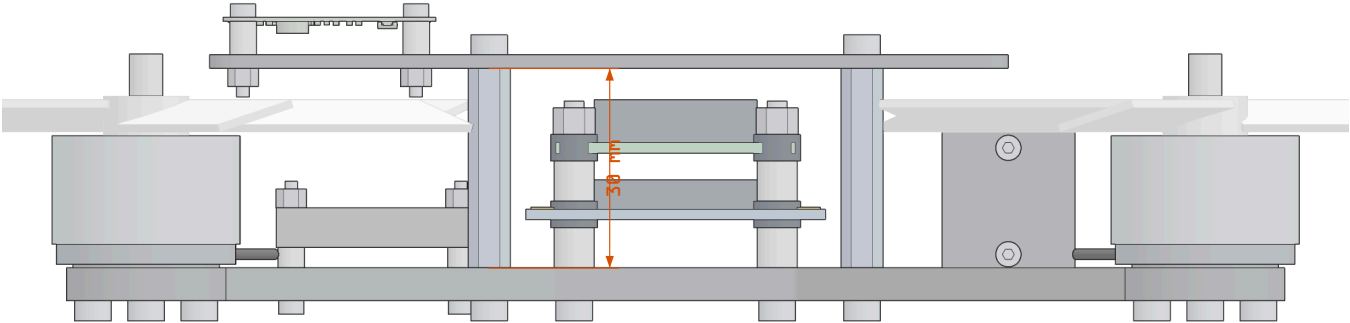


Figure 3 Right side, front to the right. Plate gap is the frame's standoff height. (unit: mm)

MASS BUDGET

PART	QTY	EACH	BASIS
CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	1	211 g	stated
MEPS NEON 2207 V2 1950KV Brushless Motor	4	36 g	stated
5-inch quadcopter frame	1	91 g ^A	CAD × 3K woven carbon fibre / epoxy laminate
5-inch frame top plate	1	19.8 g ^A	CAD × 3K woven carbon fibre / epoxy laminate
HQProp Ethix S5 5x4x3 Propeller	4	3.7 g	stated
M3 × 8 mm motor screw	16	0.91 g ^A	CAD × A2 stainless steel
Matek M9N-5883 GNSS & Compass	1	14.5 g	stated
DolphinRC AM32 60A 4-in-1 ESC	1	13.2 g	stated
M3 × 30 mm socket head cap screw	4	2.3 g ^A	CAD × A2 stainless steel
HGLRC Kevlar battery strap 20x250mm	1	8.3 g	stated
DJI O4 Air Unit	1	8.2 g	stated
DolphinRC F405 V3 Flight Controller	1	7.7 g	stated
M3 × 8 mm socket head cap screw	8	0.91 g ^A	CAD × A2 stainless steel
M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	4	1.82 g	stated
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	4	0.61 g ^A	CAD × A2 stainless steel
M2 × 18 mm socket head cap screw	4	0.57 g ^A	CAD × A2 stainless steel
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	1	2.2 g	stated
M2 nyloc nut, DIN 985, A2 stainless	8	0.24 g ^A	CAD × A2 stainless steel
M2 × 12 mm socket head cap screw	4	0.43 g ^A	CAD × A2 stainless steel
ETTINGER Ø05.83.060 M3 round spacer 6 mm, PA6 GF25	8	0.15 g	stated
M2 × 5 mm camera screw	4	0.27 g ^A	CAD × A2 stainless steel
M2 nylon spacer, Ø2.7/4.0 × 3 mm	4	0.05 g	stated
M2 nylon spacer, Ø2.7/4.0 × 5 mm	4	0.05 g	stated

All-up weight 584.0 g^A (systemMass), of which 151.2 g^A is computed from assumed material or nominal CAD geometry. Not included, no mass in the model: FC/ESC SH1.0 8-pin cable ×1, Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor ×1, XT60 battery lead (12 AWG) ×1, DJI O4 3-in-1 cable ×1. Rows in order of total mass (quantity × each).

INTERFACE GEOMETRY

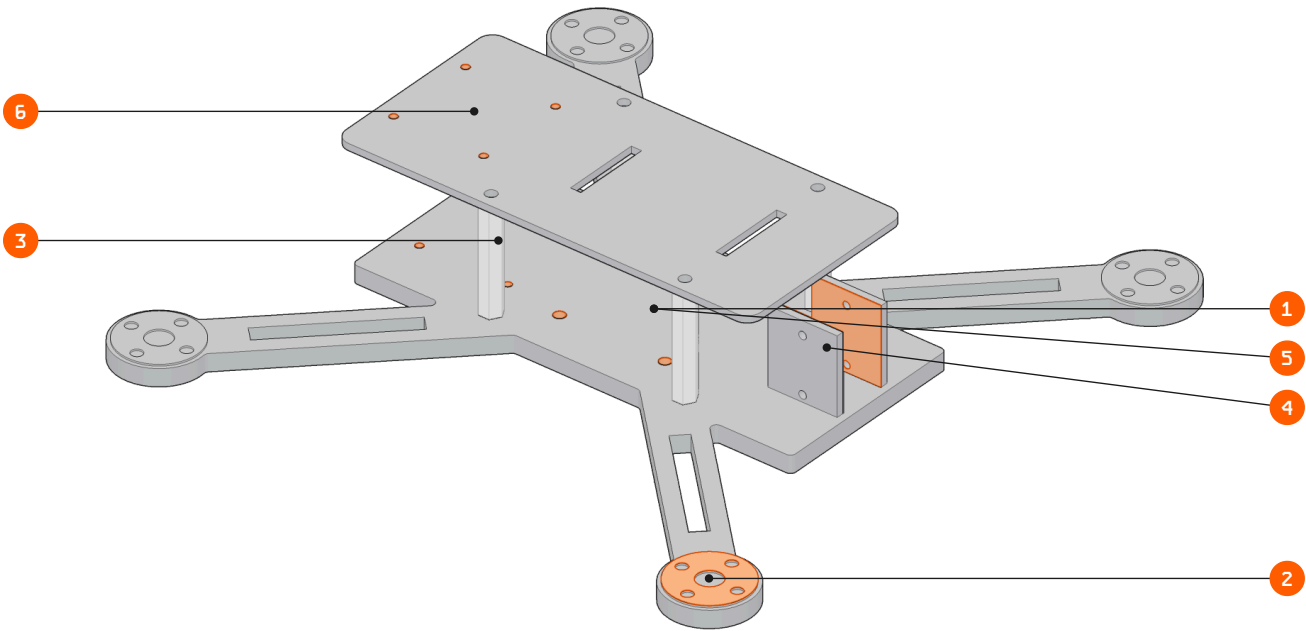


Figure 4 Structural interfaces of the frame and top plate (accent), from each interface's geometry frame and CAD feature binding.

#	INTERFACE	PATTERN	FASTENER	MATES WITH	BINDING
1	Stack mount frame.stack_mount	4 × on 30.5 mm square	3 mm	DolphinRC AM32 60A 4-in-1 ESC via stack_hardware DolphinRC F405 V3 Flight Controller via stack_hardware	D1 D2 D3
2	Front-right motor mount frame.motor_mount_fr	4 × on Ø16 mm circle	3 mm	MEPS NEON 2207 V2 1950KV Brushless Motor via arm_fr_hardware	D1 D2 D3
3	Transmission module mount frame.vtx_mount	4 × on 25.5 mm square	2 mm	DJI O4 Air Unit via vtx_hardware	D1 D2 D3
4	Camera side plates frame.camera_mount	4 × on 16 mm × 14 mm rectangle	2 mm	DJI O4 Air Unit via camera_hardware	D1 D2
5	Top-plate standoffs frame.standoff_mount	4 × on 56 mm × 48 mm rectangle	3 mm	5-inch frame top plate (custom) via frame_hardware	D1 D2 D3
6	GPS mount top_plate.gps_mount	4 × on 26 mm square	2 mm	Matek M9N-5883 GNSS & Compass via gps_hardware	D1 D2 D3

Binding: D1 named CAD feature, D2 per-interface artifact, D3 procedural from parameters (docs/geometry-artifacts.md).

04 INTERFACES AND PINOUT

Allocation of the DolphinRC F405 V3 Flight Controller (stack/fc) interfaces, pad to pad, from the stored links. Crossovers (TX→RX) are marked ✖. The link state is the result of pairwise DRC.

HUB INTERFACE	PADS (HUB → PEER)	PEER	PROTOCOL	PARAMETERS	STATE
UART2	RX2 → TX ✖ TX2 → RX ✖	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver CRSF (UART)	UART	baud rate 420 000 Bd	CONFIGURED
5V BEC	5V → 5V	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver 5V	POWER	voltage 5 V max current 3 A	CONFIGURED
GND	GND → G	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver G	POWER	—	CONFIGURED
10V/12V BEC	10V → VCC	DJI 04 Air Unit VCC (3-in-1 cable, red)	POWER	voltage 3.7–13.2 V max current 2.5 A	CONFIGURED
GND	GND → GND	DJI 04 Air Unit GND — power (3-in-1 cable, black)	POWER	—	CONFIGURED
UART5	R5 → TX ✖ TX5 → RX ✖	DJI 04 Air Unit UART (MSP DisplayPort / Canvas OSD)	UART	baud rate 115 200 Bd voltage 3.3 V	CONFIGURED
UART1	RX1 → TX ✖ TX1 → RX ✖	Matek M9N-5883 GNSS & Compass GNSS UART (NEO-M9N)	UART	baud rate 38 400 Bd	CONFIGURED
I2C1	SDA → DA SCL → CL	Matek M9N-5883 GNSS & Compass Compass I2C (QMC5883P / QMC5883L)	I2C	clock freq 100–400 kHz i2c address 0x2C	CONFIGURED
4.5V	4.5V → 5V	Matek M9N-5883 GNSS & Compass 5V	POWER	voltage 5 V	CONFIGURED
GND	GND → G	Matek M9N-5883 GNSS & Compass G	POWER	—	CONFIGURED

05 ELECTRICAL CHARACTERISTICS

BEC and rail utilisation (known loads)



Figure 5 checkSystem supply_budget.

Propulsion peak current vs battery ratings

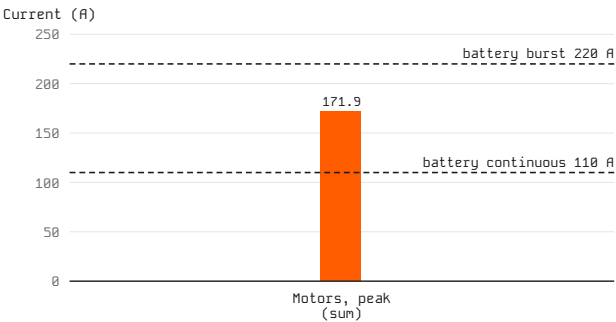


Figure 6 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.

POWER RAILS

RAIL	VOLTAGE	CAPACITY	KNOWN LOAD	LOADS	UNBUDGETED
5V BEC stack:bec_5v	5.0 V	15 W	0.225 W	receiver:vcc_5v, gnss:vin_5v, stack:rail_4v5	receiver:vcc_5v
10V/12V BEC stack:bec_10v	10.0 V	25 W	10 W	video:vcc	—
4.5V stack:rail_4v5	4.5–5.0 V	from stack:bec_5v	in parent A	stack:bec_5v, gnss:vin_5v	budgeted on stack:bec_5v (assumed branch)

From checkSystem rule supply_budget: capacity = rail voltage * current rating; loads that state no draw cannot be budgeted. A branch rail (supplied_from) is budgeted on the output that feeds it; A marks a relation assumed, not cited.

SUPPLY INPUTS

MODULE	INPUT	VOLTAGE RANGE	MIN. SUPPLY POWER
DolphinRC F405 V3 Flight Controller	VBAT VBAT	6.0–30.0 V	—
DolphinRC AM32 60A 4-in-1 ESC	BAT+ +	6.0–30.0 V	—
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	5V 5V	5.0 V	—
DJI O4 Air Unit	VCC (3-in-1 cable, red) VCC	3.7–13.2 V	10 W
Matek M9N-5883 GNSS & Compass	5V 5V	5.0 V	—
Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	+ lead +	0.0–35.0 V	—

Stored links: 28; configured: 28. Diagnostics on the nominal build: warning propulsion_current, info supply_budget, info supply_budget, info supply_budget, info prop_handedness, info fastener_torque.

06 PERFORMANCE

TEST DATA

MOTOR THRUST TABLES (MANUFACTURER)

THROTTLE	SZ4942 V	A	RPM	G	W	SZ5145 V	A	RPM	G	W
10 %	25.2	0.3	4888	28	9	25.2	0.4	4854	33	9
20 %	25.2	1.3	9570	119	33	25.2	1.4	9465	143	35
30 %	25.1	3.3	14 157	277	83	25.1	3.6	13 911	334	90
40 %	25.1	5.8	17 710	452	145	25.0	6.4	17 304	533	159
50 %	25.0	9.5	20 929	648	237	24.9	10.4	20 457	763	260
60 %	24.9	14.3	23 893	865	355	24.8	15.5	23 263	1000	384
70 %	24.7	20.0	26 534	1083	494	24.7	21.5	25 704	1233	531
80 %	24.5	27.3	28 890	1298	668	24.5	28.7	27 751	1446	701
90 %	24.3	35.2	30 766	1470	855	24.3	36.8	29 199	1580	893
100 %	24.2	41.8	31 973	1584	1009	24.1	43.0	30 342	1699	1036

MEPS NEON 2207 V2 1950KV Brushless Motor: every row of the maker's KV table, per test propeller (MEPS SZ4942 at 25.2 V; MEPS SZ5145 at 25.2 V); columns per propeller are measured supply voltage, current, speed, thrust and electrical power. The fitted propeller (HQProp Ethix S5 5x4x3 Propeller) is not one the maker tested, so these figures, and every figure derived from them below, are for the maker's test props.

CAUTION Charts and figures tagged **STAND-IN • NOT MEASURED** are derived from the model to show the expected shape. Replace them by dropping measured files into `library/systems/quadcopter-5in/docs/test-data/` (status `measured`); the next build swaps them in.

Thrust and current per motor vs throttle

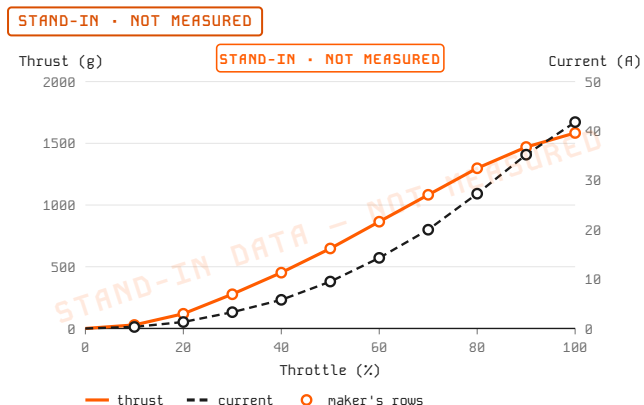


Figure 7 Circles: every row of the maker's table (MEPS SZ4942 at 25.2 V, not the fitted prop). Lines: Stand-in: the motor maker's thrust table row by row, scaled to all motors, plus avionics draw.

Flight time by regime

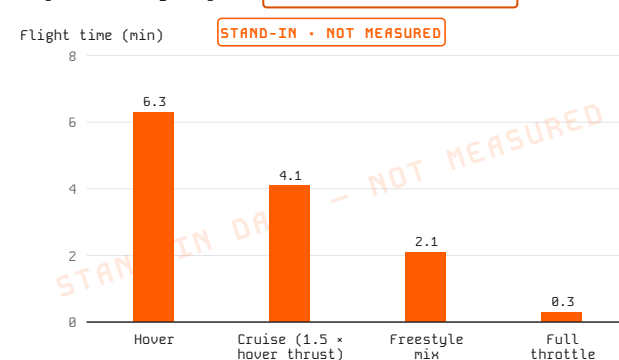


Figure 8 Stand-in: time = capacity × usable fraction ÷ average current. Static-thrust currents: pessimistic for forward flight.

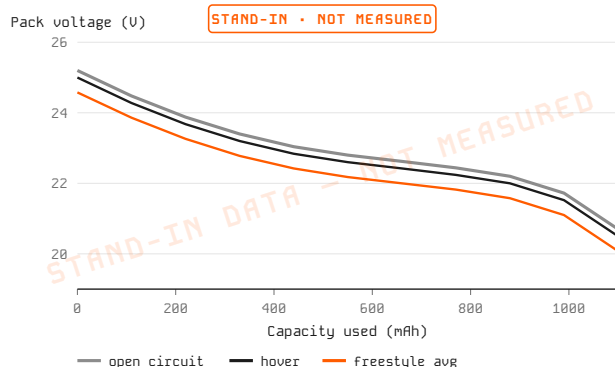
Battery discharge **STAND-IN • NOT MEASURED**

Figure 9 Stand-in: generic LiPo open-circuit curve per cell minus $I \cdot R$ sag, scaled to the pack.

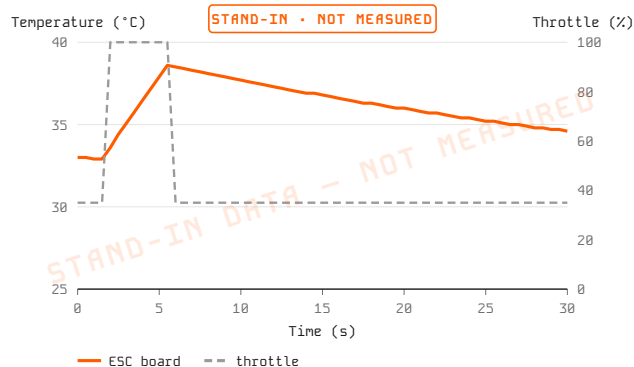
ESC temperature over a punch-out **STAND-IN • NOT MEASURED**

Figure 10 Stand-in: lumped thermal model $C \cdot dT/dt = P - (T - \text{ambient})/R$ with conduction and switching losses per channel.

PERFORMANCE SUMMARY

FIGURE	VALUE	TEST DATA
All-up weight (estimate)	604 g ^S	hover-current
Hover current	8.4 A ^S	hover-current
Hover power	188 W ^S	hover-current
Avionics current	0.7 A ^S	hover-current
Total static thrust at 100 %	6.34 kg ^S	throttle-sweep
Total current at 100 %	167.9 A ^S	throttle-sweep
Thrust-to-weight at 25.2 V ^S	10.5 ^S : 1	thrust-to-weight
Thrust-to-weight at 22.2 V ^S	8.1 ^S : 1	thrust-to-weight
Thrust-to-weight at 21.0 V ^S	7.3 ^S : 1	thrust-to-weight
Flight time, hover	6.3 min ^S	flight-time
Flight time, freestyle mix	2.1 min ^S	flight-time
ESC peak temperature	38.6 °C ^S	esc-temp-punchout

STAND-IN ASSUMPTIONS

- motor figures interpolate linearly between the 10 rows of the maker's table (0 at 0 % throttle); pack current = table electrical power ÷ 22.2 V nominal
- maker's table used the MEPS S24942 prop at 25.2 V, not the fitted propeller
- avionics: DJI O4 minimum supply power 10 W, GNSS 50 mA at 5 V, FC + receiver 2.5 W (assumed), BEC efficiency 85 % (assumed)
- all-up weight = 584 g from systemMass (derived:mass.all_up, of which 151.2 g is CAD volume × assumed material) + 20 g assumed for the 4 parts with no mass in the model (FC/ESC SH1.0 8-pin cable, Panasonic EEU-FR1U471 470 µF 35 V low-ESR electrolytic capacitor, XT60 battery lead (12 AWG), DJI O4 3-in-1 cable)
- usable capacity 80 % of rated
- thermal resistance 2 K/W and capacity 20 J/K (board in prop wash) – not in the model
- effective conduction resistance 4 m ohm per channel and 1.5 W switching loss per channel at full throttle – not in the model
- starting 8 °C above ambient after the previous flight segment
- generic LiPo open-circuit voltage curve
- internal resistance 4 m ohm per cell – not in the model

07 DESIGN CHECKS

CHECKSYSTEM

The nominal build and each fault scenario in `scenarios.ts` run through `checkSystem`. A fault scenario is expected to raise at least one error or warning.

SCENARIO	DIAGNOSTICS	FINDINGS (ERRORS AND WARNINGS)
Nominal build nominal	0 ERR 5 INFO	PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
04 wired to raw 6S o4-on-ubatt	2 ERR 4 INFO	LINK_STATE 04 10V: incompatible – voltage: 18–25.2 V vs 3.7–13.2 V (ranges do not overlap) HARNESS_CONNECTOR dji_cable.end_fc is dji_6pin but battery:battery_out is xt60. PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
20 × 20 frame for a 30.5 stack frame-20mm-stack	2 ERR 5 INFO	LINK_STATE ESC on frame: incompatible – hole_spacing: 30.5 mm vs 20 mm (ranges do not overlap); fastener_diameter: 3 mm vs 2 mm (ranges do not overlap) LINK_STATE FC on frame: incompatible – hole_spacing: 30.5 mm vs 20 mm (ranges do not overlap); fastener_diameter: 3 mm vs 2 mm (ranges do not overlap) PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
GPS on the receiver's UART gps-on-rx-uart	1 ERR 5 INFO	INTERFACE_REUSE stack:uart2 is linked 2 times (rx_crsf, gnss_uart) but supports 1. PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
Legacy GNSS (M10Q-5883, EOL) legacy-m10q	0 ERR 5 INFO	PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
One prop on backwards prop-reversed	1 ERR 5 INFO	PROP_HANDEDNESS arm_fr/prop_on_shaft: CW prop on a motor spinning CCW (arm_fr/motor); it would push air the wrong way. Fit the CCW prop. PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
Two compasses on one I2C bus two-compasses	1 ERR 5 INFO	BUS_ADDRESS stack:i2c1: 2 devices share I2C address 0x2C (gnss:i2c_compass, gnss_2:i2c_compass). PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.

08 BILL OF MATERIALS

GENERATED/BOM.JSON

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
1	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) cnhl-black-series-1100mah-6s-100c	CNHL (China Hobby Line) 1101006BK	1	PURCHASE	211 g	audited 18/18 A 3 · gaps 7
2	DJI 04 Air Unit dji-04-air-unit	DJI DF3L2904	1	PURCHASE	8.2 g	audited 32/34 A 1 · gaps 5
3	DolphinRC AM32 60A 4-in-1 ESC dolphinrc-am32-60a-4in1-esc	DolphinRC (Shenzhen DolphinRC Model Technology Co., Ltd.) DolphinRC AM32 ESC 60A (F405 V3 stack, SKU DPD200041 option '60A ESC')	1	PURCHASE	13.2 g	audited 18/18 A 1 · gaps 5
4	DolphinRC F405 V3 Flight Controller dolphinrc-f405-v3-flight-controller	DolphinRC (Shenzhen DolphinRC Model Technology Co., Ltd.) DolphinRC F405 V3 (F405 V3 stack, SKU DPD200041 option 'F405-FC')	1	PURCHASE	7.7 g	audited 28/30 A 1 · gaps 8
5	ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 ettinger-005-83-060-m3-nylon-spacer-6mm	ETTINGER 005.83.060	8	PURCHASE	0.15 g	audited 12/12 A 1 · gaps 1
6	HGLRC Kevlar battery strap 20x250mm hglrc-kevlar-battery-strap-20x250mm	HGLRC F.20.02.00010	1	PURCHASE	8.3 g	audited 12/12 A 1 · gaps 5
7	HQProp Ethix S5 5x4x3 Propeller hqprop-ethix-s5	HQProp EthixS5-GR-PC	4 pack of 4	PURCHASE	3.7 g	audited 14/14 A 0 · gaps 4
8	M2 nyloc nut, DIN 985, A2 stainless iso-10511-m2-nyloc-nut	Westfield Fasteners WF16826	8	PURCHASE	0.24 g ^A	— A 1 · gaps 0
9	M2 nylon spacer, Ø2.7/4.0 x 3 mm m2-nylon-spacer-3mm	Ettinger 007.02.403	4 pack of 100	PURCHASE	0.05 g	— A 1 · gaps 0
10	M2 nylon spacer, Ø2.7/4.0 x 5 mm m2-nylon-spacer-5mm	Ettinger 007.02.405	4 pack of 100	PURCHASE	0.05 g	— A 1 · gaps 0
11	M2 x 12 mm socket head cap screw, ISO 4762 iso-4762-m2x12	Generic (ISO 4762 / DIN 912) WF2332	4	PURCHASE	0.43 g ^A	— A 1 · gaps 0
12	M2 x 18 mm socket head cap screw, ISO 4762 iso-4762-m2x18	Generic (ISO 4762 / DIN 912) WF17972	4	PURCHASE	0.57 g ^A	— A 1 · gaps 0
13	M3 aluminium standoff, 30 mm, female-female, 5.5 mm Af m3-aluminium-standoff-30mm	Ettinger 005.03.308	4 pack of 100	PURCHASE	1.82 g	— A 1 · gaps 0
14	M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless iso-10511-m3-nyloc-nut	Generic (ISO 10511 / DIN 985) WF1288	4	PURCHASE	0.61 g ^A	audited 12/12 A 1 · gaps 2
15	M3 x 30 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m3x30-socket-head-cap-screw	Generic (ISO 4762 / DIN 912) WF2352	4	PURCHASE	2.3 g ^A	audited 12/12 A 1 · gaps 2
16	M3 x 8 mm socket head cap screw, ISO 4762 iso-4762-m3x8	Generic (ISO 4762 / DIN 912) WF2344	8	PURCHASE	0.91 g ^A	— A 1 · gaps 0
17	Matek M9N-5883 GNSS & Compass matek-m9n-5883	Matek Systems M9N-5883	1	PURCHASE	14.5 g	audited 26/26 A 2 · gaps 8

Rows as generated by scripts/bom.ts (generated/bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
18	MEPS NEON 2207 V2 1950KV Brushless Motor meps-neon-2207-v2-1950kv	MEPS NEON 2207 V2 1950KV	4	PURCHASE	36 g	audited 20/20 A 0 · gaps 7
19	Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor panasonic-eeu-fr1v471	Panasonic Industry EEU-FR1V471	1	PURCHASE	—	audited 17/18 A 2 · gaps 3
20	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver radiomaster-rp1-v2-elrs-2g4	RadioMaster RP1 V2	1	PURCHASE	2.2 g	audited 21/21 A 0 · gaps 4
21	5-inch frame top plate (custom) quadcopter-5in-top-plate	Delta Robotics (custom) QUAD5-TOP	1	FABRICATE	19.8 g ^A	— A 1 · gaps 0
22	5-inch quadcopter frame (custom) quadcopter-5in-frame	Delta Robotics (custom) QUAD5-FRAME	1	FABRICATE	91 g ^A	— A 1 · gaps 0
23	DJI O4 3-in-1 cable dji-o4-3in1-cable	DJI	1	INCL. dji-o4-air-unit	—	— A 0 · gaps 0
24	FC/ESC SH1.0 8-pin cable dolphinrc-sh8-fc-esc-cable	DolphinRC	1	INCL. dolphinrc-f405-v3-flight-controller	—	— A 0 · gaps 0
25	M2 x 5 mm camera screw (included with the O4) o4-camera-screw-m2x5	Included (unbranded)	4	INCL. dji-o4-air-unit	0.27 g ^A	— A 1 · gaps 1
26	M3 x 8 mm motor screw (included with the motor) motor-screw-m3x8	Included (unbranded)	16	INCL. meps-neon-2207-v2-1950kv	0.91 g ^A	— A 1 · gaps 1
27	XT60 battery lead (12 AWG) dolphinrc-xt60-battery-lead	DolphinRC	1	INCL. dolphinrc-am32-60a-4in1-esc	—	— A 0 · gaps 0

Rows as generated by scripts/bom.ts (generated/bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

09 EVIDENCE AND ASSUMPTIONS

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	5	18/18	—	minimum (discharge cut-off) cell voltage; operating / charging / storage temperature; burst duration; main lead and balance lead length; XT60 gender (+2)
DJI O4 Air Unit	5	32/34	—	antenna connector family; FC-side 3-in-1 cable connector family; power consumption / max input current; storage temperature; lens mount in box
DolphinRC AM32 60A 4-in-1 ESC	3	18/18	—	pin 1 orientation of 8-pin SH connector (silkscreen order only); burst current duration (5 s vs 10 s) (sources conflict); operating temperature (not stated); VBAT passthrough current limit on connector V pin (not stated); current sensor output voltage range (only Betaflight scale/offset given)
DolphinRC F405 V3 Flight Controller	4	28/30	A 4.5V rail supply: branch of bec_5v (5 V 3 A) via a diode	pin 1 orientation of all SH connectors (Manual shows silkscreen order only.); which UART the SBUS pad / DJI Sbus pin feeds, and inversion (Not stated; Betaflight target defaults serial RX to UART2.); 'ppm' silkscreen near RX3 pad (Low-res photo; target puts PPM on UART2 RX (PA3). Not claimed.); S1/S2 pad function (Product page says 10x PWM outputs (8 DShot), target defines SERVO1-3; S1/S2 modelled as PWM outputs only.); RSSI/Airspeed ADC pads (Listed as ADC channels but no pad labelled RSSI/Airspeed in drawings; omitted.) (+3)
ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25	3	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommeted board clamp height 4.0 mm each (ESC and FC)	operating temperature
HGLRC Kevlar battery strap 20x250mm	3	12/12	—	strap thickness; tensile strength / load rating; buckle dimensions and anti-slip strip length; temperature range; frame top-plate thickness and slot width
HQProp Ethix S5 5x4x3 Propeller	2	14/14	—	hub mounting style / retention; recommended motor / cell count; operating temperature; thrust / efficiency data
M2 nyloc nut, DIN 985, A2 stainless	—	not audited	A part choice: M2 nyloc nut, DIN 985, A2 stainless	—
M2 nylon spacer, Ø2.7/4.0 x 3 mm	—	not audited	A part choice: M2 nylon spacer, Ø2.7/4.0 x 3 mm	—
M2 nylon spacer, Ø2.7/4.0 x 5 mm	—	not audited	A part choice: M2 nylon spacer, Ø2.7/4.0 x 5 mm	—
M2 x 12 mm socket head cap screw, ISO 4762	—	not audited	A length: 12 mm	—
M2 x 18 mm socket head cap screw, ISO 4762	—	not audited	A length: 18 mm	—
M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF	—	not audited	A part choice: M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	2	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommets board clamp height 4.0 mm each (ESC and FC)	mass; nylon insert temperature limit
M3 x 30 mm socket head cap screw, ISO 4762, A2 stainless	2	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommets board clamp height 4.0 mm each (ESC and FC)	mass; operating temperature
M3 x 8 mm socket head cap screw, ISO 4762	—	not audited	A length: 8 mm	—
Matek M9N-5883 GNSS & Compass	13	26/26	—	pin 1 of JST-GH-6P; edge pads VBus, D-, D+, Bt and V-USB jumper; fastener size for 2 mm holes; hole pattern Y spacing; logic levels, I2C pull-ups on module (+3)
MEPS NEON 2207 V2 1950KV Brushless Motor	5	20/20	—	lead wire gauge (AWG); continuous current rating; operating / storage temperature; maximum voltage (V); prop nut thread handedness / shaft rotation (+2)
Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	3	17/18	A identity: Panasonic EEU-FR1V471 (Ø10 x 16 mm) A lead polarity identification: negative lead marked by the sleeve stripe	bundled capacitor brand/series/MPN; mass; ESR (as such)
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	8	21/21	—	supply current; operating temperature; logic level of TX/RX pads; receiver sensitivity (RP1-specific)
5-inch frame top plate (custom)	—	not audited	A material: 3K woven carbon fibre, 1.52 g/cm ³	—
5-inch quadcopter frame (custom)	—	not audited	A material: 3K woven carbon fibre, 1.52 g/cm ³	—
DJI O4 3-in-1 cable	—	not audited	—	—
FC/ESC SH1.0 8-pin cable	—	not audited	—	—
M2 x 5 mm camera screw (included with the O4)	—	not audited	A length: 5 mm	screw length
M3 x 8 mm motor screw (included with the motor)	—	not audited	A length: 8 mm	screw length
XT60 battery lead (12 AWG)	—	not audited	—	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

10 VALUE PROVENANCE

This document shows 514 bound values. Each is a query into the UHD model, re-resolved and compared by `uhd-tech-docs-verify`; the full list with sources is in `uhd-quad5-ds.values.json` next to the document.

STATUS	VALUES	MEANING
Cited source	174	stated by a source in the part's sources.json
Model design value	267	a design value of a custom module, or a UHD check result
Derived	26	computed from model values; formula in the provenance appendix
Assumption	18	the model marks this value as an assumption (assumption trait)
Stand-in	15	stand-in data derived from the model, not measured
Not in model	14	the model does not state this value (data gap)

STATUS	VALUE	QUERY	DERIVATION / REASON
D Derived	6.3 kg	derived:propulsion.thrust_full_total	motors × thrust_g of the top (100 %) row of the first thrust table · manufacturer table with MEPS S24942 at 25.2 V (not the fitted propeller) · inputs: 1
A Assumption	584 g	derived:mass.all_up	systemMass: $\sum$ quantity × (weight_g, else CAD volume × material density) over every instance; instances with neither are listed, not guessed · excludes 4 part(s) with no mass: FC/ESC SH1.0 8-pin cable, Panasonic EEU-FR1V471 470 μ F 35 V low-ESR electrolytic capa
A Assumption	91.0 g	derived:mass.module(quadcopter-5in-frame)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 59.86 cm ³ × 1.52 g/cm ³ (3K woven carbon fibre / epoxy laminate); assumption: DragonPlate's woven quasi-isotropic row is taken for 3K woven plate; the t
D Derived	24.42 Wh	derived:power.battery_energy_Wh	battery voltage × capacity · inputs: 2
A Assumption	bec_5v	def:dolphinrc-f405-v3-flight-controller:interfaces[id=rail_4v5].traits[type=supplied_from].params.interfaceId	assumption trait · https://dolphinrc.com/u_file/2608/24/file/DolphinRCF40520250616153927-165929bb5a.pdf
D Derived	159.1 mm	derived:frameDistance(frame.motor_mount_fl,frame.motor_mount_fr)	placement(a) · a.frame.origin - placement(b) · b.frame.origin from assemble() · inputs: 2
D Derived	159.1 mm	derived:frameDistance(frame.motor_mount_rr,frame.motor_mount_fr)	placement(a) · a.frame.origin - placement(b) · b.frame.origin from assemble() · inputs: 2
D Derived	159.1 mm	derived:frameDistance(frame.motor_mount_fl,frame.motor_mount_rl)	placement(a) · a.frame.origin - placement(b) · b.frame.origin from assemble() · inputs: 2
A Assumption	3K woven carbon fibre / epoxy laminate	def:quadcopter-5in-frame:domains[domain=mechanical].material.name	assumption trait · https://dragonplate.com/carbon-fiber-sheets
A Assumption	1.52 g/cm ³	def:quadcopter-5in-frame:domains[domain=mechanical].material.density_g_cm3	assumption trait · https://dragonplate.com/carbon-fiber-sheets
A Assumption	19.8 g	derived:mass.module(quadcopter-5in-top-plate)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 13.05 cm ³ × 1.52 g/cm ³ (3K woven carbon fibre / epoxy laminate); assumption: DragonPlate's woven quasi-isotropic row is taken for 3K woven plate; the t
A Assumption	0.91 g	derived:mass.module(motor-screw-m3x8)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.11 cm ³ × 8 g/cm ³ (A2 (304 / 1.4301) stainless steel); assumption: The maker does not state the screw material; steel (A2 density) and the generator's

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
A Assumption	2.3 g	derived:mass.module(iso-4762-m3x30-socket-head-cap-screw)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.28 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: The CAD body is library/cad/py/parts.py's nominal ISO 4762 shape (plain shank, no t
A Assumption	0.91 g	derived:mass.module(iso-4762-m3x8)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.11 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	0.61 g	derived:mass.module(iso-10511-m3-nyloc-nut)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.08 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: The CAD body is library/cad/py/parts.py's nominal DIN 985 hex with the nylon insert
A Assumption	0.57 g	derived:mass.module(iso-4762-m2x18)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.07 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	0.24 g	derived:mass.module(iso-10511-m2-nyloc-nut)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.03 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Nominal hex nut shape with the nylon insert counted as steel (a slight over-estimat
A Assumption	0.43 g	derived:mass.module(iso-4762-m2x12)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.05 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	0.27 g	derived:mass.module(o4-camera-screw-m2x5)	weight_g if stated, else CAD body volume (generator manifest) * domains[mechanical].material.density_g_cm3 * 0.03 cm³ * 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: The maker does not state the screw material; steel (A2 density) and the generator's
A Assumption	151.2 g	derived:mass.assumed	Σ systemMass entries whose material carries an assumption * inputs: 10
S Stand-in	604 g	test: hover-current:summary[id=auw_g].value	library/systems/quadcopter-5in/docs/test-data/hover-current.json
S Stand-in	8.4 A	test: hover-current:summary[id=hover_current_A].value	library/systems/quadcopter-5in/docs/test-data/hover-current.json
S Stand-in	188 W	test: hover-current:summary[id=hover_power_W].value	library/systems/quadcopter-5in/docs/test-data/hover-current.json
S Stand-in	0.7 A	test: hover-current:summary[id=avionics_current_A].value	library/systems/quadcopter-5in/docs/test-data/hover-current.json
S Stand-in	6.34 kg	test: throttle-sweep:summary[id=thrust_total_full].value	library/systems/quadcopter-5in/docs/test-data/throttle-sweep.json
S Stand-in	167.9 A	test: throttle-sweep:summary[id=current_total_full].value	library/systems/quadcopter-5in/docs/test-data/throttle-sweep.json
S Stand-in	25.2 V	test: thrust-to-weight:rows[0][0]	library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json
S Stand-in	10.5	test: thrust-to-weight:rows[0][2]	library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
S Stand-in	22.2 V	test:thrust-to-weight:rows[1][0]	library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json
S Stand-in	8.1	test:thrust-to-weight:rows[1][2]	library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json
S Stand-in	21.0 V	test:thrust-to-weight:rows[2][0]	library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json
S Stand-in	7.3	test:thrust-to-weight:rows[2][2]	library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json
S Stand-in	6.3 min	test:flight-time:summary[id=hover_min].value	library/systems/quadcopter-5in/docs/test-data/flight-time.json
S Stand-in	2.1 min	test:flight-time:summary[id=mix_min].value	library/systems/quadcopter-5in/docs/test-data/flight-time.json
S Stand-in	38.6 °C	test:esc-temp-punchout:summary[id=peak_C].value	library/systems/quadcopter-5in/docs/test-data/esc-temp-punchout.json
D Derived	0	derived:diagCount(nominal,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(nominal,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	5	derived:diagCount(nominal,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	2	derived:diagCount(o4-on-vbat,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(o4-on-vbat,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	4	derived:diagCount(o4-on-vbat,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	2	derived:diagCount(frame-20mm-stack,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(frame-20mm-stack,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	5	derived:diagCount(frame-20mm-stack,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(gps-on-rx-uart,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(gps-on-rx-uart,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	5	derived:diagCount(gps-on-rx-uart,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	0	derived:diagCount(legacy-m10q,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(legacy-m10q,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	5	derived:diagCount(legacy-m10q,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(prop-reversed,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(prop-reversed,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	5	derived:diagCount(prop-reversed,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(two-compasses,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(two-compasses,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	5	derived:diagCount(two-compasses,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
- Not in model	-	sys:bom[line=8].verification	UHD bom
- Not in model	-	sys:bom[line=9].verification	UHD bom
- Not in model	-	sys:bom[line=10].verification	UHD bom
- Not in model	-	sys:bom[line=11].verification	UHD bom
- Not in model	-	sys:bom[line=12].verification	UHD bom
- Not in model	-	sys:bom[line=13].verification	UHD bom
- Not in model	-	sys:bom[line=16].verification	UHD bom
- Not in model	-	sys:bom[line=21].verification	UHD bom
- Not in model	-	sys:bom[line=22].verification	UHD bom
- Not in model	-	sys:bom[line=23].verification	UHD bom
- Not in model	-	sys:bom[line=24].verification	UHD bom
- Not in model	-	sys:bom[line=25].verification	UHD bom
- Not in model	-	sys:bom[line=26].verification	UHD bom
- Not in model	-	sys:bom[line=27].verification	UHD bom